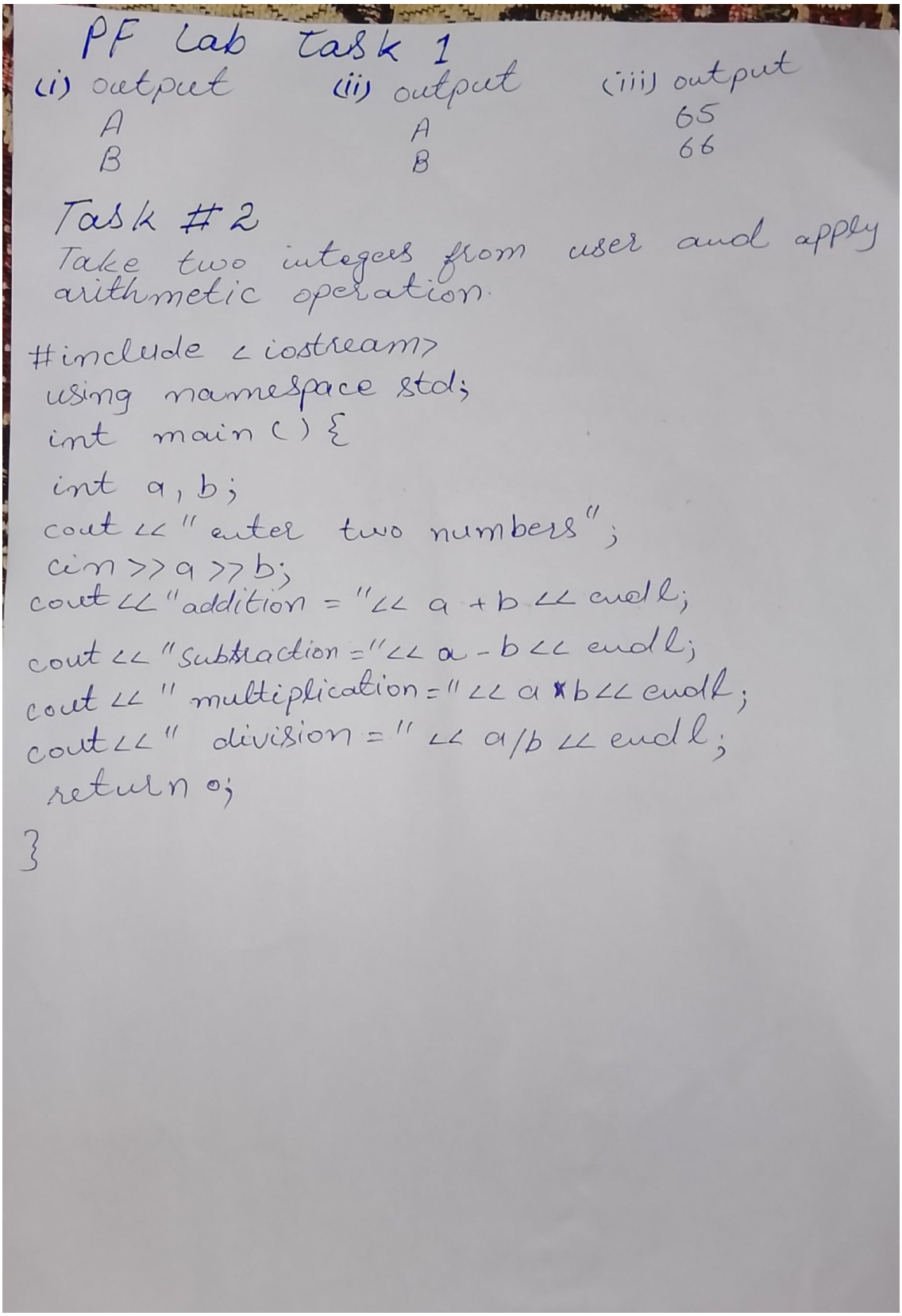
Programming Fundamentals – Lab #02

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Tasks 1 & 2 (handwritten)



PF Lab Task 1

(i) output
A
B

(ii) output
A
B

(iii) output
65
66

Task #2

Take two integers from user and apply arithmetic operation.

```
#include <iostream>
using namespace std;
int main() {
int a, b;
cout << "enter two numbers";
cin >> a >> b;
cout << "addition = " << a + b << endl;
cout << "subtraction =" << a - b << endl;
cout << " multiplication =" << a * b << endl;
cout << " division = " << a/b << endl;
return 0;
}
```

Task 2 – Code and Output

Code

```cpp
#include <iostream>
using namespace std;
int main()
{
    int a, b;
    cout << "Enter two numbers: ";
    cin >> a >> b;
    cout << a << " + " << b << " = " << a + b << endl;
    cout << a << " - " << b << " = " << a - b << endl;
    cout << a << " * " << b << " = " << a * b << endl;
    cout << a << " / " << b << " = " << a / b << endl;
    return 0;
}
```

Tasks 3 & 4 (handwritten)

Task # 3

output if we enter 5, 10, 15

5
10
15
15
10
5

Task # 4

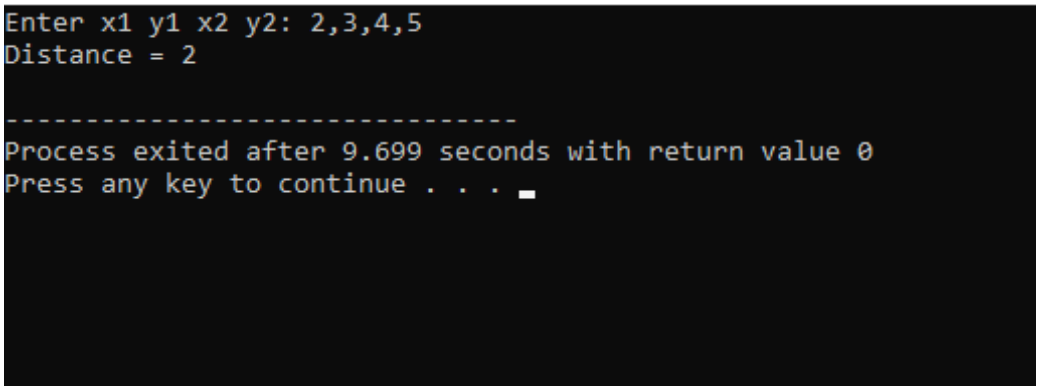
```
#include <iostream>
using namespace std;
#include <cmath>
using namespace std;
int main(){
double x1, y1, x2, y2, d;
cout << "enter x1 y1 x2 y2";
cin >> x1 >> y1 >> x2 >> y2;
d = sqrt(pow(x2-x1,2) + pow(y2-y1,2));
cout << "distance = " << d << endl;
return 0;
}
```

Task 4 – Code and Output

Code

```cpp
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
    double x1, y1, x2, y2, d;
    cout << "Enter x1 y1 x2 y2: ";
    cin >> x1 >> y1 >> x2 >> y2;
    d = sqrt(pow(x2 - x1, 2) + pow(y2 - y1, 2));
    cout << "Distance = " << d << endl;
    return 0;
}
```

Output



Task 5 (handwritten)

Task #5

calculate area (Pi x r^2) circle
area of rectangle (length x width)
area of triangle (base x height x 0.5)

```
#include <stdio.h
#include <iostream>
using namespace std;
  int main (){
  const double PI = 3.14159;
  double r, length, width, base, height,
  cout << "enter radius of circle ";
   cin >> r;
   cout << "area of circle = " << PI * r * r << endl
   cout << "enter length and width of
          rectangle" ;
  cin >> length >> width;
  cout << "area of rectangle = " << length * wid
       << endl;
  cout << "enter base and height of triang
   cin >> base >> height ;
   cout << "area of triangle = " << base *
        height * 0.5 << endl;
        return 0;
    }
```

Task 5 – Code and Output

Code

```cpp
#include <iostream>
using namespace std;

int main()
{
    const double PI = 3.14159;
    double r, length, width, base, height;

    cout << "Enter radius of circle: ";
    cin >> r;
    cout << "Area of Circle = " << PI * r * r << endl;

    cout << "Enter length and width of rectangle: ";
    cin >> length >> width;
    cout << "Area of Rectangle = " << length * width << endl;

    cout << "Enter base and height of triangle: ";
    cin >> base >> height;
    cout << "Area of Triangle = " << base * height * 0.5 << endl;

    return 0;
}
```

Output

```
Enter radius of circle: 5
Area of Circle = 78.5397
Enter length and width of rectangle: 12,5
Area of Rectangle = 0
Enter base and height of triangle: Area of Triangle = nan

--------------------------------
Process exited after 16.56 seconds with return value 0
Press any key to continue . . .
```

Task 6 (handwritten)

Task #6

```
#include <iostream>
#include <iomanip>
using namespace std;
int main(){
double item1=12.95, item2=24.95, item 3=6
    item4=14.95, item5=3.95;
const double tax-rate=0.06;
double subtotal = item1+item2+item3+item
                  item5;
double tax = subtotal * tax-rate;
double total = subtotal + tax;
cout<< fixed << setprecision(2);
cout<<"price of item1 = $" << item1 << endl;
cout<<"price of item2 = $" << item2 << endl;
cout<<"price of item 3 = $" << item 3 << endl;
cout<<"price of item 4 = $" << item4 << endl;
cout<<"price of item 5 = $" << item5 << endl;
cout<<"subtotal = $" << subtotal << endl;
cout<<"sales tax = $" << tax << endl;
cout<<"total = $" << total << endl;
    return 0;
}
```

Task 6 – Code and Output

Code

```cpp
#include <iostream>
#include <iomanip>
using namespace std;

int main()
{
    double item1 = 12.95, item2 = 24.95, item3 = 6.95, item4 = 14.95, item5 = 3.95;
    const double TAX_RATE = 0.06;

    double subtotal = item1 + item2 + item3 + item4 + item5;
    double tax = subtotal * TAX_RATE;
    double total = subtotal + tax;

    cout << fixed << setprecision(2);
    cout << "Price of item 1: $" << item1 << endl;
    cout << "Price of item 2: $" << item2 << endl;
    cout << "Price of item 3: $" << item3 << endl;
    cout << "Price of item 4: $" << item4 << endl;
    cout << "Price of item 5: $" << item5 << endl;
    cout << "Subtotal:        $" << subtotal << endl;
    cout << "Sales tax (6%):  $" << tax << endl;
    cout << "Total:           $" << total << endl;

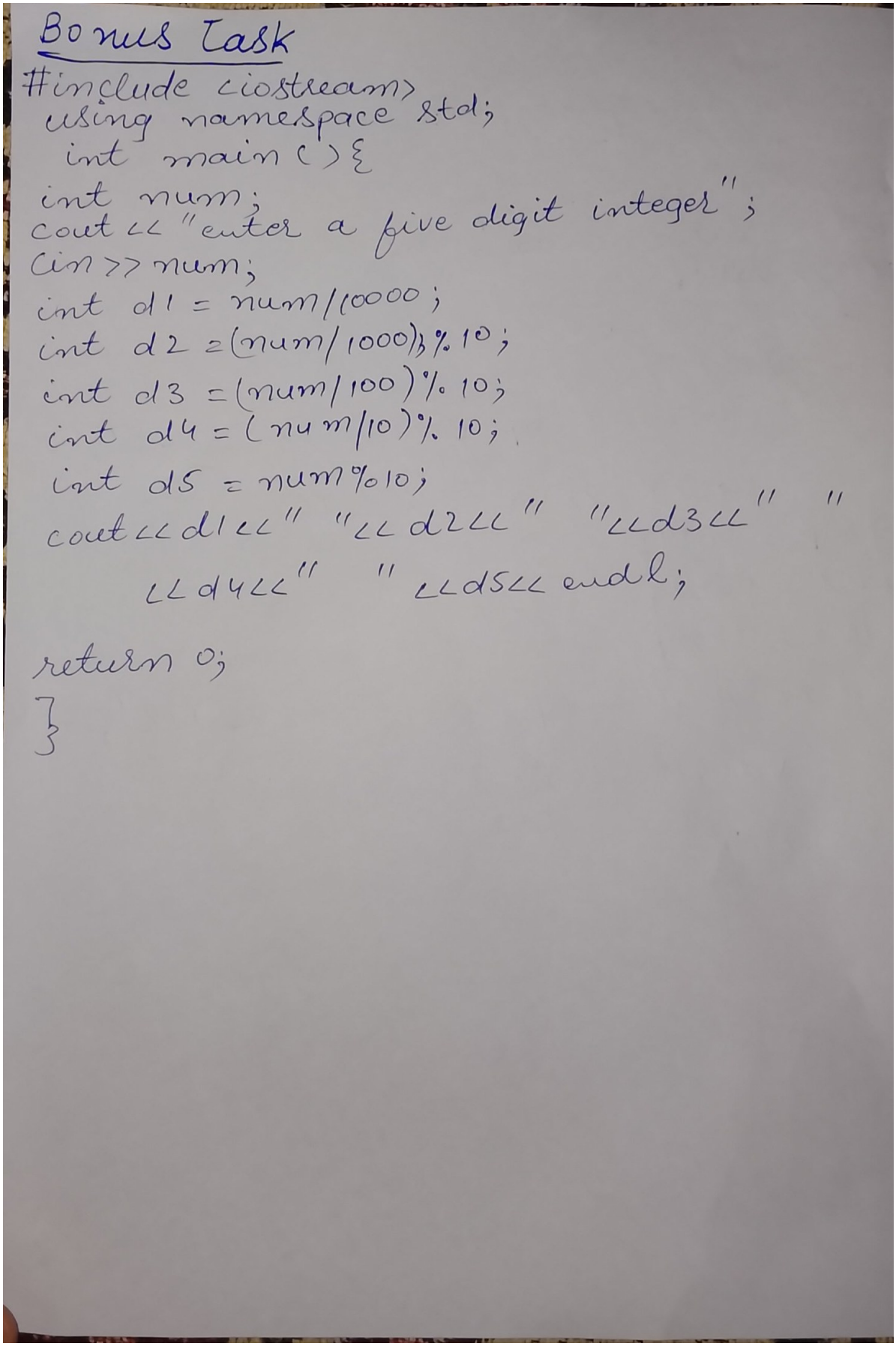
    return 0;
}
```

Output

```
Price of item 1: $12.95
Price of item 2: $24.95
Price of item 3: $6.95
Price of item 4: $14.95
Price of item 5: $3.95
Subtotal:        $63.75
Sales tax (6%):  $3.82
Total:           $67.58

--------------------------------
Process exited after 0.3979 seconds with return value 0
Press any key to continue . . .
```

Bonus Task (handwritten)



Bonus Task

```
#include <iostream>
using namespace std;
int main(){
int num;
cout << "enter a five digit integer";
cin >> num;
int d1 = num/10000;
int d2 = (num/1000)%10;
int d3 = (num/100)%10;
int d4 = (num/10)%10;
int d5 = num%10;
cout << d1 << " " << d2 << " " << d3 << " "
     << d4 << " " << d5 << endl;

return 0;
}
```

Bonus Task – Code and Output

Code

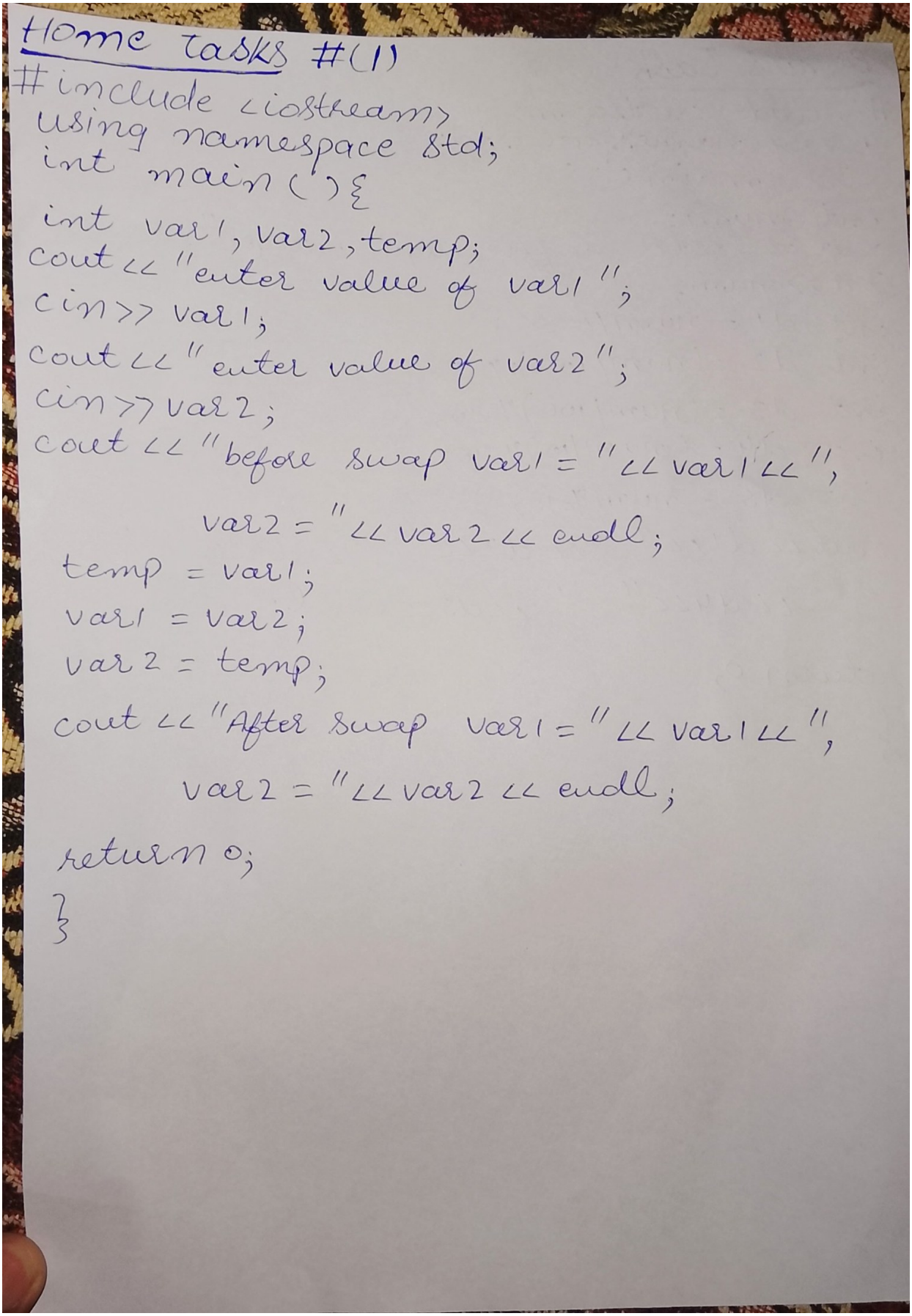
```cpp
#include <iostream>
using namespace std;
int main()
{
    int num;
    cout << "Enter a five-digit integer: ";
    cin >> num;
    int d1 = num / 10000;
    int d2 = (num / 1000) % 10;
    int d3 = (num / 100) % 10;
    int d4 = (num / 10) % 10;
    int d5 = num % 10;
    cout << d1 << "   " << d2 << "   " << d3 << "   " << d4 << "   " << d5 << endl;
    return 0;
}
```

Output

```
Enter a five-digit integer: 12345
1   2   3   4   5

--------------------------------
Process exited after 9.732 seconds with return value 0
Press any key to continue . . .
```

Home Task 1 (handwritten)

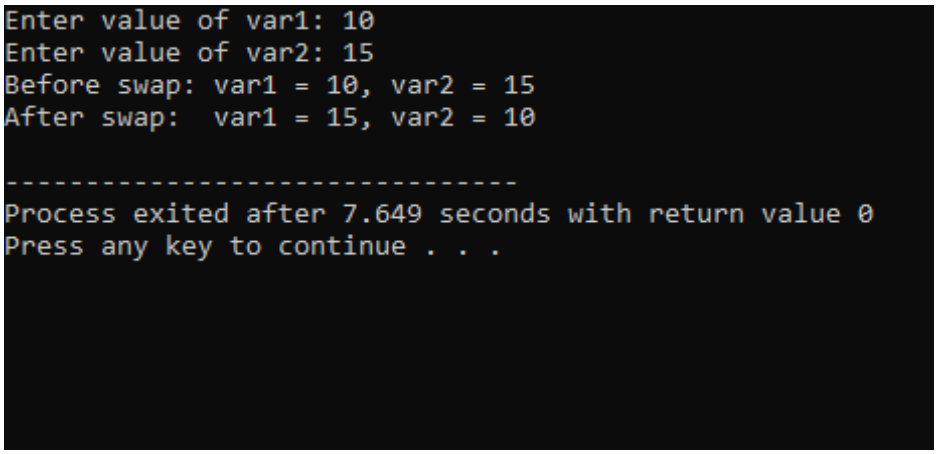


Home Task 1 – Code and Output

Code

```cpp
#include <iostream>
using namespace std;
int main()
{
    int var1, var2, temp;
    cout << "Enter value of var1: ";
    cin >> var1;
    cout << "Enter value of var2: ";
    cin >> var2;
    cout << "Before swap: var1 = " << var1 << ", var2 = " << var2 << endl;
    temp = var1;
    var1 = var2;
    var2 = temp;
    cout << "After swap:  var1 = " << var1 << ", var2 = " << var2 << endl;
    return 0;
}
```

Output



Home Task 2 (handwritten)

Home task #2

```
#include <iostream>
using namespace std;
int main ( ){
char ch;
cout << "enter an alphabet in small
        caps";
cin >> ch;
 char upper = ch-32;
 cout << "uppercase" << endl upper;
 return 0;
}
```

Home Task 2 – Code and Output

Code

```cpp
#include<iostream>
using namespace std;

int main() {
    char ch;
    cout << "Enter an alphabet in small caps: ";
    cin >> ch;

    // Convert using ASCII
    char upper = ch - 32;

    cout << "Uppercase: " << upper;

    return 0;
}
```

Output

```
Enter an alphabet in small caps:
```